BRAKE MASTER CYLINDER PUSHROD REPLACEMENT

AFFECTED VEHICLES: **CERTAIN 2002 MODEL YEAR TH!NK *neighbor***

REQUIRED SERVICE TOOLS

1/4 in. drive ratchet
1/4 in. drive deep-well 10 mm socket
1/4 in. drive 2 in. extension
13 mm wrench
1/4 in. drive click-type torque wrench
Diagonal wire cutters
Pliers
Flashlight
Telescoping magnet

SERVICE PROCEDURE

1. Apply the parking brake.
2. **WARNING: REMOVE ALL JEWELRY AND WATCHES BEFORE PERFORMING THE FOLLOWING STEP. USE CAUTION WHEN WORKING ON OR AROUND THE BATTERIES.**

   WARNING: THE BATTERY PACK ASSEMBLY CAN DELIVER IN EXCESS OF 72 VOLTS OF DC POWER. IMPROPER HANDLING OF THE BATTERY PACK CAN RESULT IN INJURY OR FATALITY. ONLY AUTHORIZED PERSONNEL TRAINED TO WORK WITH BATTERY PACK COMPONENTS ARE PERMITTED TO HANDLE THE BATTERIES.

   Shut off the main power switch.
   - Remove the driver seat cushion.
   - Reach under the seat area and move the switch to the OFF position. See Figure 1.

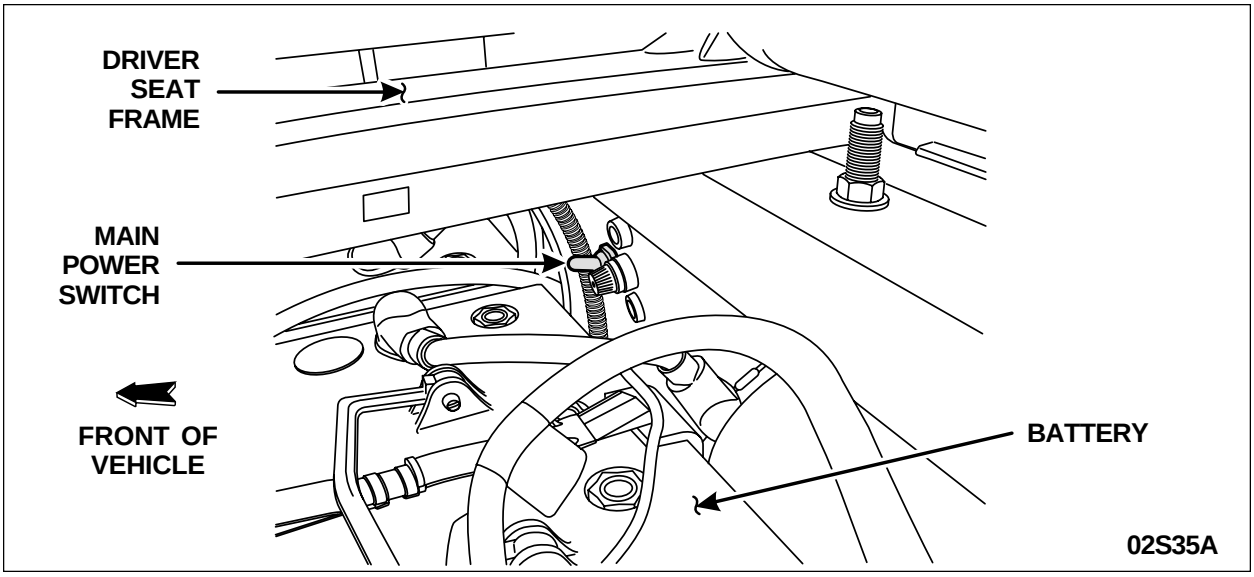


FIGURE 1

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3. Remove the hood.
4. Remove the two (2) bolts and two (2) nuts securing the brake master cylinder to its mounting bracket. Then position the brake master cylinder out of the way. See Figure 2.

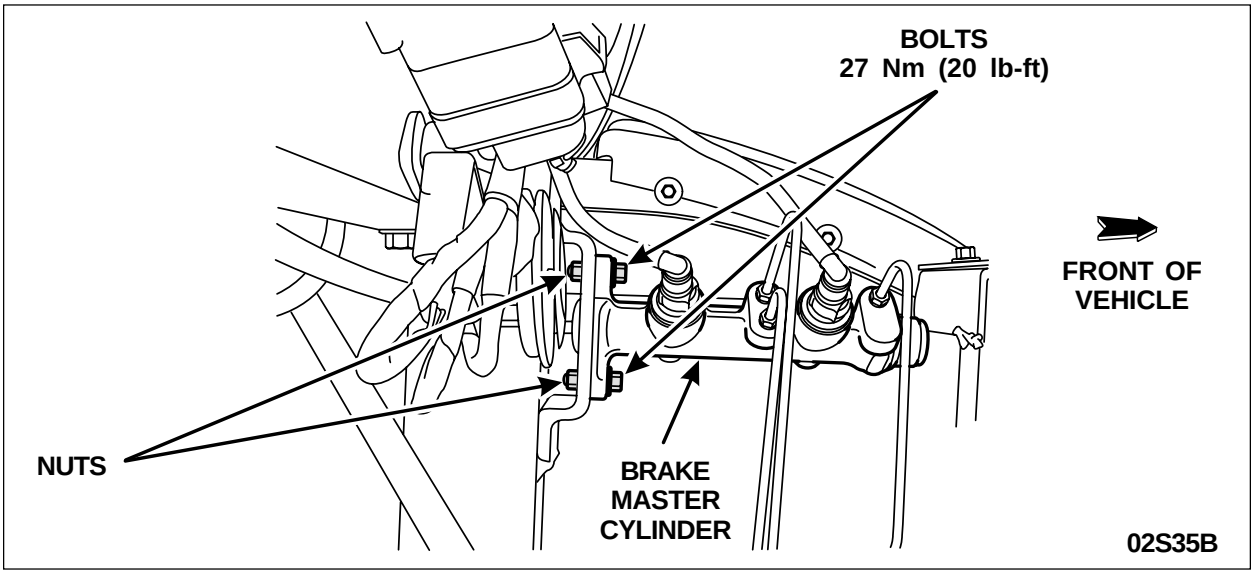


FIGURE 2

5. **CAUTION: Be careful not to damage any of the wire harnesses or electrical relays while removing and installing the cotter pin and/or pushrod assembly.**

   Remove the cotter pin and the clevis pin from the pushrod clevis. See Figure 3.

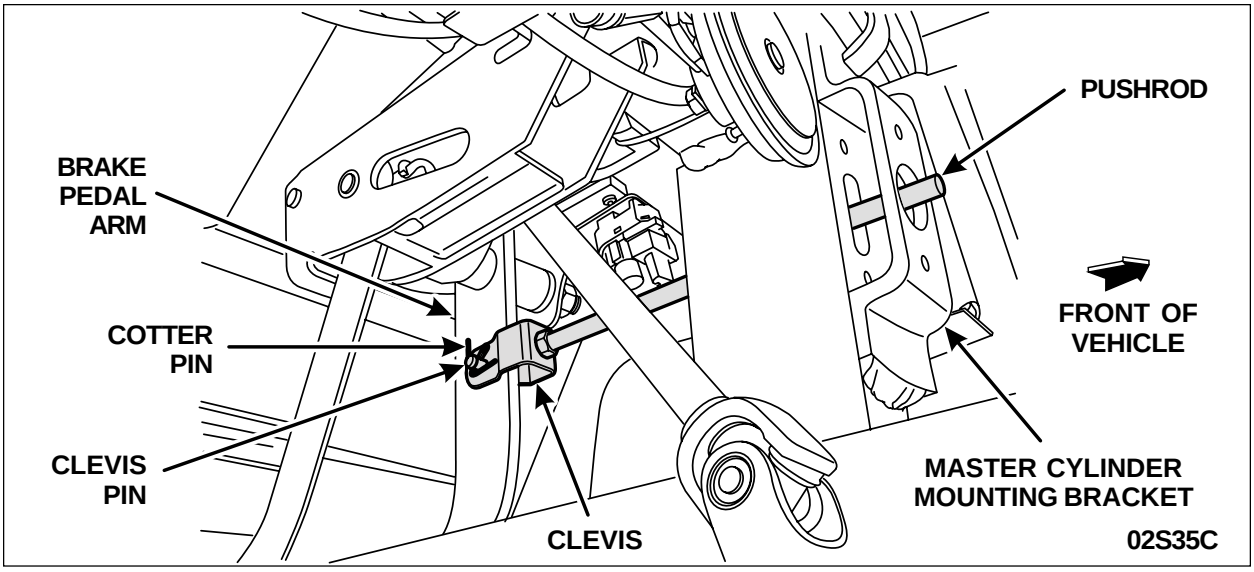


FIGURE 3

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6. NOTE: To ease pushrod assembly removal and installation, it may be necessary to position the brake pedal arm slightly to the side.

   Remove the pushrod assembly from the vehicle pulling the pushrod out along the passenger side of the brake pedal arm and toward the bulkhead.

7. NOTE: The ***new*** pushrod assembly comes preadjusted. As long as the jam nut and pushrod are not disturbed, there is no need for pushrod adjustment.

   Position the ***new*** pushrod assembly into the vehicle.

8. Position the pushrod assembly against the brake pedal arm and install the ***new*** clevis pin. Install a ***new*** cotter pin into the clevis pin.

9. NOTE: Verify that the black dust cap is still in place on the brake master cylinder before positioning it back onto its mounting bracket.

   Position the brake master cylinder back onto its mounting bracket, making sure to seat the end of the pushrod into the master cylinder. Then install the two (2) nuts and the (2) bolts. Tighten the fasteners to 27 Nm (20 lb-ft).

10. Apply the brake pedal three (3) times and then check the master cylinder brake line connections for leaks.

11. Install the hood.

12. **WARNING: REMOVE ALL JEWELRY AND WATCHES BEFORE PERFORMING THE FOLLOWING STEP. USE CAUTION WHEN WORKING ON OR AROUND THE BATTERIES.**

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    Turn on the main power switch.
    - Reach under the seat area and move the main power switch to the ON position.
    - Install the driver seat cushion.

13. Release the parking brake.

14. Verify brake and brake lamp operation.

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15. **WARNING: THERE ARE GASES AROUND THE BATTERIES THAT CAN EXPLODE IF EXPOSED TO FLAMES, SPARKS, OR LIT CIGARETTES. THE AMOUNT OF GAS IS INCREASED DURING BATTERY CHARGING. AN EXPLOSION COULD RESULT IN PERSONAL INJURY OR VEHICLE DAMAGE.**

    WARNING: BATTERIES CONTAIN SULFURIC ACID, WHICH CAN BURN SKIN, EYES, AND CLOTHING, IF CONTACTED.

    WARNING: DO NOT CHARGE THE BATTERIES WITH THE WEATHER ENCLOSURE CLOSED OR THE VEHICLE COVER IN PLACE. A BUILD UP OF HYDROGEN GAS CAN RESULT WHICH CAN EXPLODE. THE CHARGING AREA SHOULD BE WELL VENTILATED.

    CAUTION: If the vehicle is allowed to sit in conditions of - 6° C (20° F) or less with a charge of 20% (two bars showing on gauge) or less, the batteries could freeze. Allowing the batteries to freeze may cause permanent damage to the batteries and permanently reduce their capacity. In cold conditions, place the vehicle in an area greater than 0° C (32° F) and allow it to warm up before charging. Never charge the vehicle if the batteries may be frozen. Allow the batteries to warm above 0° C (32° F) first, then charge.

    CAUTION: Do not park and leave the vehicle with discharged batteries. The batteries could discharge to the point where damage could occur and the battery charger will not charge.

    NOTE: The batteries could require up to 12 hours of charge. The charging system is equipped with an automatic shut-off feature to prevent over-charging of the batteries.

    a) Retrieve the ground fault circuit interrupt (GFCI) charger cord from the vehicle's trunk.
    b) Turn the Mode Selector key to the OFF position, remove the key and set the parking brake.
    c) NOTE: Make sure the power point, if equipped, is not in use and the hazard lamps are not in use.

       NOTE: Do not use any plug adapters or extension cords to make the electrical connection.

       Plug the GFCI charger cord into the wall outlet, then open the charge inlet access door, located to the left of the steering wheel, and fully insert the GFCI charger cord plug into it.
    d) Verify the POWER indicator is illuminated on the GFCI charger cord. Also, the battery charge indicator on the instrument cluster will illuminate during the charging process.
    e) Once the batteries are fully charged, push the TEST button on the GFCI charger cord and verify the power indicator on the GFCI is off.
    f) Disconnect the GFCI charger cord from the wall outlet and then from the charge inlet and close the charge inlet access door.
    g) Return the GFCI charger cord to the vehicle's trunk.

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